5.6 Power Series

Definition 5.6.1 (Convergent Power Series)
*Let K be a field. Then a **power series** is a formal expression of the form*

$$P(X) \coloneqq \sum_{n=0}^{\infty} a_n X^n$$

where $a_n \in K$. We denote the ring of power series by $K[[X]]$. If K is a valued field then, we say this converges at $z \in K$ if the series

$$\sum_{n=0}^{\infty} a_n z^n$$

converges in K.

Proposition 5.6.2 (Radius of convergence)
Let K be a complete valued field and consider the formal power series

$$\sum_{n=0}^{\infty} a_n X^n \quad a_n \in K \ .$$

*Define the **radius of convergence** to be*

$$R \coloneqq \left(\limsup \sqrt[n]{|a_n|}\right)^{-1}$$

where we understand $0^{-1} = \infty$ and $\infty^{-1} = 0$. Then

a) For every $|z| < R$ the series $\sum_{n=0}^{\infty} a_n z^n$ is absolutely convergent, and we denote this function by $f(z)$.
b) For every $|z| > R$ the series $\sum_{n=0}^{\infty} a_n z^n$ is divergent
c) R is the unique value such that a), b) hold.
d) The limit $f(z)$ is continuous and differentiable in the open set $B(0,R)$ with derivative

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

which has radius of convergence at least R

Proof.

a) First consider the case $R < \infty$. Define $c_n \coloneqq a_n z^n$ then $\sqrt[n]{|c_n|} = |z| \sqrt[n]{|a_n|}$. Then clearly $|z| < R \implies \limsup \sqrt[n]{|c_n|} < 1$ whence the result follows from (5.4.31). In the case $R = \infty$ we see that $\limsup \sqrt[n]{|c_n|} = 0$ for all z and the result follows similarly.
b) This follows similarly by (5.4.27).
c) This is evident for suppose $R' \neq R$ satisfies the same properties and consider z such that $|z| = \frac{R'+R}{2}$ for a contradiction.
d) Let R' be the radius of convergence for the given series, we claim that $R' \geq R$, or in otherwords that $|z| < R$ implies $\sum_{n=0}^{\infty}(n+1)a_{n+1}z^n$ converges. For we may choose w such that $|z| < |w| < R$. Then by definition

$$\sum_{n=0}^{\infty} a_n w^n$$

converges. In particular by (...) $|a_n w^n| \to 0$ and therefore $|a_n w^n|$ is bounded, by M. Therefore

$$\sum_{n=1}^{\infty} \left|n a_n z^{n-1}\right| = \sum_{n=1}^{\infty} n \left|a_n w^{n-1}\right| \left|\frac{z}{w}\right|^{n-1} \leq \frac{M}{|w|} \sum_{n=1}^{\infty} n \left|\frac{z}{w}\right|^{n-1}$$

The latter series converges by the Ratio Test (5.4.32) and therefore the series converges by the Comparison Test (5.4.28). Consider distinct $z, w \in B(0,R)$ then

$$\frac{f(w) - f(z)}{w - z} - f'(z) = \sum_{n=2}^{\infty} a_n \left[\frac{w^n - z^n}{w - z} - n z^{n-1}\right]$$

Observe that for $n \geq 1$

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$$\frac{w^n - z^n}{w - z} = w^{n-1} + w^{n-2}z + \cdots + wz^{n-2} + z^{n-1}$$

Therefore for $n \geq 2$

$$\begin{aligned}\frac{w^n - z^n}{w - z} - nz^{n-1} &= \big(w^{n-1} - z^{n-1}\big) + z\big(w^{n-2} - z^{n-2}\big) + \cdots + z^{n-2}(w - z) \\ &= (w - z)\big[w^{n-2} + 2w^{n-3}z + \cdots + (n-1)z^{n-2}\big]\end{aligned}$$

For any w such that $|w - z| < \frac{R-|z|}{2}$ we have $|w| \leq \frac{R+|z|}{2} =: r < R$. Then

$$\left|\frac{w^n - z^n}{w - z} - nz^{n-1}\right| \leq |w - z|\frac{n(n-1)}{2}r^{n-2}$$

which implies

$$\left|\frac{f(w) - f(z)}{w - z} - f'(z)\right| \leq |w - z|\sum_{n=2}^{\infty}|a_n|\frac{n(n-1)}{2}r^{n-2}$$

The sum is finite because the second formal derivative is absolutely convergent at r, by the part already demonstrated. Therefore we see that f is differentiable at z with derivative $f'(z)$. This also shows that f is continuous at z (...).

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